

# MLCA

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```
## -----  
##' [libraries]  
## -----
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.1.5  
v forcats    1.0.1      v stringr    1.5.2  
v ggplot2    4.0.0      v tibble     3.3.0  
v lubridate  1.9.4      v tidyr      1.3.1  
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

```
library(tidycensus)
```

```
library(multilevLCA)
```

multilevLCA: Estimates and plots single- and multilevel latent class models.

```
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
library(tigris)
```

To enable caching of data, set ``options(tigris_use_cache = TRUE)`` in your R script or `.Rprofile`.

```
library(nnet)
library(VGAM)
```

Loading required package: stats4  
Loading required package: splines

Attaching package: 'VGAM'

The following object is masked from 'package:psych':

logistic

```
library(gt)
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:psych':

logit

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
library(writexl)
```

```
rm(list = ls())
```

```
## -----
##' [IPEDS/Carnegie Classification Data]
## -----
```

```
# load IPEDS data and merge with the Carnegie Classification
```

```
ipeds24 <- read_csv("../data/sources/hd2024.csv")
```

```
Rows: 6072 Columns: 72
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (23): INSTNM, IALIAS, ADDR, CITY, STABBR, ZIP, CHFM, CHFTITLE, EIN, UEI...
```

```
dbl (49): UNITID, FIPS, OBEREG, GENTLE, OPEFLAG, SECTOR, ICLEVEL, CONTROL, ...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ipeds_cost <- read_csv("../data/sources/cost1_2024.csv")
```

```
Rows: 5867 Columns: 418
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (198): XROOMCAP, XMEALSWK, XAPPFEEU, XAPPFEEG, XAPPFEED, FAFQ2URL, XROOM...
```

```
dbl (214): UNITID, ROOM, ROOMCAP, ALLONCAM, TUITVARY, BOARD, MEALSWK, APPLFE...
```

```
lgl (6): CHG10AY0, CHG10AY1, CHG10AY2, CHG10PY0, CHG10PY1, CHG10PY2
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#ipeds_schoolsize <- read_csv("../data/sources/ef2024a.csv") #enrollment Fall 2024
carnegie <- read_excel("../data/sources/2025-Public-Data-File.xlsx", sheet = 2)
ic2024 <- read_csv("../data/sources/ic2024.csv")
```

```
Rows: 5963 Columns: 111
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): XDISABPC
```

```
dbl (110): UNITID, PE01ISTR, PE02ISTR, PE03ISTR, PE04ISTR, PE05ISTR, PE06IST...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
sfa <- read_csv("../data/sources/sfa2324.csv")
```

Rows: 5575 Columns: 183

-- Column specification -----

Delimiter: ","

chr (91): XSCUGRAD, XSCUGDGSK, XSCUGNDGS, XSCUGFFN, XSCUGFFP, XSCFA2, XSCFA2...

dbl (92): UNITID, SCUGRAD, SCUGDGSK, SCUGNDGS, SCUGFFN, SCUGFFP, SCFA2, SCFA...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
adm <- read_csv("../data/sources/adm2024.csv")
```

Rows: 1956 Columns: 101

-- Column specification -----

Delimiter: ","

chr (44): XAPPLCN, XAPPLCNM, XAPPLCNW, XAPPLCNAN, XAPPLCNUN, XADMSSN, XADMSS...

dbl (57): UNITID, ADMCON1, ADMCON2, ADMCON3, ADMCON4, ADMCON5, ADMCON6, ADMC...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
enroll <- read_csv("../data/sources/effy2024.csv") %>%
  filter(EFFYALEV == 3) %>% #degree/certificate-seeking undergraduates
  select(UNITID, EFFYALEV, EFYTOTLT, EFYAIANT, EFYBKAAT, EFYNHPIT, EFYWHITT, EFYHISPT, EFYAS...
```

Rows: 115026 Columns: 72

-- Column specification -----

Delimiter: ","

chr (34): XEYTOTLT, XEYTOTLM, XEYTOTLW, XEFYAIAT, XEFYAIAM, XEFYIAIW, XEFYAS...

dbl (38): UNITID, EFFYALEV, EFFYLEV, LSTUDY, EFYTOTLT, EFYTOTLM, EFYTOTLW, E...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
#ipeds_distance <- read_csv("../data/sources/effy2024_dist.csv") %>%
# filter(EFFYDLEV == 2) %>% #undergrad total
# select(UNITID, EFFYDLEV, EFYDETOT, EFYDEEXC)
ipeds_dist <- read_csv("../data/sources/ef2024a_dist.csv") |>
  filter(EFDELEV == 2) |> #undergrad total
  select(UNITID, EFDELEV, EFDETOT, EFDEEXC, XEFDEEXC)
```

Rows: 21819 Columns: 20

-- Column specification -----

Delimiter: ","

chr (9): XEFDETOT, XEFDEEXC, XEFDESOM, XEFDENON, XEFDEEX1, XEFDEEX2, XEFDEE...

dbl (11): UNITID, EFDELEV, EFDETOT, EFDEEXC, EFDESOM, EFDENON, EFDEEX1, EFDE...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
ipeds_learner <- read_csv("../data/sources/ef2024b.csv") |>
  filter(LSTUDY == 2, EFBAGE %in% c(1, 7)) |>      #undergrad total / all age categories total
#number of institutions (= UNITID): 2,487
  select(UNITID, EFBAGE, EFAGE06, EFAGE09) |>
  pivot_wider(names_from = EFBAGE,
              values_from = c(EFAGE06, EFAGE09))
```

Rows: 76359 Columns: 22

-- Column specification -----

Delimiter: ","

chr (9): XEFAGE01, XEFAGE02, XEFAGE03, XEFAGE04, XEFAGE05, XEFAGE06, XEFAGE...

dbl (13): UNITID, EFBAGE, LINE, LSTUDY, EFAGE01, EFAGE02, EFAGE03, EFAGE04, ...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
#Let's replace this with the size variable in IPEDS (instsize)
# ipeds_schoolsize <- ipeds_schoolsize |>
#   filter(EFALEVEL == 1) |> #Level of student (1 == all students total)
#   select(UNITID, EFTOTLT) |> #Grand total
#   mutate(size = case_when(
#     EFTOTLT <= 3000 ~ 0, #small
#     EFTOTLT <= 10000 ~ 1, #medium
#     EFTOTLT > 10000 ~ 2, #large
#     TRUE ~ NA_real_))

carnegie <- carnegie |>
  select(unitid, hsi)

# merge files
data <- ipeds24 %>%
  left_join(carnegie, by = c("UNITID" = "unitid")) %>%
  left_join(ipeds_cost, by = "UNITID") |>
```

```

#left_join(ipeds_schoolsize, by = "UNITID") %>%
left_join(ic2024, by = "UNITID" ) %>%
left_join(sfa, by = "UNITID") %>%
left_join(adm, by = "UNITID") %>%
left_join(enroll, by = "UNITID") |>
left_join(ipeds_dist, by = "UNITID") |>
left_join(ipeds_learner, by = "UNITID")

# select variables and create a dataset for analysis
x <- data |>
  rename_all(tolower) |> #converts all column names in a data frame to lowercase
  select(unitid, instnm, city, stabbr, fips, countycd, countynm, cbsa, cbsatype, csa, ober
    locale, zip, longitud, latitude,
    control, iclevel, instcat, deggrant, instsize,
    hloffer,
    carnegieic, carnegiesaec, carnegiealf, carnegieapm, carnegiersch,
    hbcu, hsi, tribal,
    room, relaffil, distnced, athassoc,
    admsn, applcn, openadmp, loan_a, tuition1, upgrntp,
    efytotlt, efyaiant, efybkaat, efyhnpit, efywhitt, efyhispt, efyasiat,
    efdelev, efdetot, efdeexc, xefdeexc, slo7, eface06_1, eface06_7, eface09_1, eface
# assign to "data"
data <- x

```

```

x <- data %>%
  filter(fips <= 56) |> # Keep only 50 states and DC
  filter(cbsa != -2) |> # Remove colleges not located in CBSA
  # filter(!is.na(control), control != 3) |> # 3: private for-profit
  filter(!is.na(iclevel), iclevel != 3) |> # 3: less than 2 years
  filter(!is.na(carnegiealf), carnegiealf != 6) %>% # 6: graduate-focused
  filter(instcat >= 2 & instcat <= 4) # 1: degree-granting, graduate with 1
  #filter(carnegieapm == 1) # 5-6: non-degree-granting
  # Keep only colleges classified as "1"

data <- x

```

```

## -----
##' [Data Cleaning]
## -----

x <- data |>
  mutate(
    public = case_when(
      control == 2 ~ 0,          # 0: private
      control == 1 ~ 1,          # 1: public
      control == 3 ~ 2,          # 2: for-profit
      TRUE ~ NA_real_),

    twoyear = case_when(
      iclevel == 1 ~ 0,          # 0: 4 or more
      iclevel == 2 ~ 1,          # 1: at least 2 & less than 4
      TRUE ~ NA_real_),

    # msi = case_when(
    #   hbcu == 1 ~ 1,            # 0: non of them
    #   hsi == 1 ~ 2,            # 1: HBCU
    #   tribal == 1 ~ 3,         # 2: HSI
    #   hbcu == 2 & hsi == 0 & tribal == 2 ~ 0,
    #   is.na(hbcu) | is.na(hsi) | is.na(tribal) ~ NA_real_,
    #   TRUE ~ NA_real_),
    #   tribal == 1 ~ 3,         # 3: TCU

    msi = case_when(
      hbcu == 1 | hsi == 1 | tribal == 1 ~ 1,
      TRUE ~ 0),

    rurality = case_when(
      locale %in% c(31,32,33,41,42,43) ~ 0,    # 0: Town + Rural
      locale %in% c(21,22,23) ~ 1,             # 1: Suburb
      locale %in% c(11,12,13) ~ 2,             # 2: City
      TRUE ~ NA_real_),

    size = case_when(
      instsize == 1 ~ 0,          # 0: < 1,000
      instsize == 2 ~ 1,          # 1: 1,000 - 4,999
      instsize == 3 ~ 2,          # 2: 5,000 - 9,999
      instsize == 4 ~ 3,          # 3: 10,000 - 19,999
      instsize == 5 ~ 4,          # 4: >=20,000
      TRUE ~ NA_real_),
  )

```

```

#       tuition = case_when(
#         tuition1 == 0 & obereg != 0 ~ NA_real_,
#         tuition1 == 0 & obereg == 0 ~ 0,
#         tuition1 < 10000 ~ 0,           # 0: < $10,000           - 1334
#         tuition1 >= 10000 & tuition1 < 20000 ~ 1,       # 1: $10,000 - $19,999 - 689
#         # tuition1 >= 20000 & tuition1 < 30000 ~ 2,       # 2: $20,000 - $29,999
#         tuition1 >= 20000 & tuition1 < 40000 ~ 2,       # 3: $30,000 - $39,999 - 531
#         # tuition1 >= 40000 & tuition1 < 50000 ~ 4,       # 4: $40,000 - $49,999
#         tuition1 >= 40000 ~ 3,           # 5: > $50,000       - 384
#         TRUE ~ NA_real_),              #                    - 316

#       tuitionnn = ntile(tuition1, 4) - 1,               # 0~3: quartiles

#check missing values: sum(is.na(ipeds_cost$TUITION1)) [1] 2,368 (= about 40% of the sample)

housing = case_when(
  room == 2 ~ 0,           # 0: no housing
  room == 1 ~ 1,           # 1: offer housing
  TRUE ~ NA_real_),

religious = ifelse(relaffil == -2, 0, 1),                 # 0: no religious
                                                         # 1: any religious

#We need to remove colleges that offer completely distance education from our sample
distance = case_when(
  distnced == 2 ~ 0,       # 0: not completely distanced learning
  distnced == 1 ~ 1,       # 1: completely distanced learning
  TRUE ~ NA_real_),

online_perc = case_when(
  is.na(efdetot) ~ NA_real_,
  is.na(efdeexc) & xefdeexc == "A" ~ 0,                 # "A": not applicable
  is.na(efdeexc) ~ NA_real_,                             # other missing cases
  TRUE ~ efdeexc / efdetot),

#       online = case_when(
#         is.na(online_perc) ~ NA_real_,
#         online_perc == 0 ~ 0,           # none - 8
#         online_perc < 0.25 ~ 1,         # more than 0%, less than 25% - 1
#         online_perc < 0.50 ~ 2,         # 25% to less than 50% - 7
#         online_perc <= 1 ~ 3,           # 50% to 100% - 2
#         TRUE ~ NA_real_),              # NA - 1

```

```

    online = ntile(online_perc, 4) - 1,                # 0~3: quartiles

#check missing values: sum(is.na(ipeds_dist$EFDETOT))  [1] 0

    evening = case_when(
      slo7 == 0 ~ 0,                                   # 0: no weekend/evening courses
      slo7 == 1 ~ 1,                                   # 1: allows classes only on weekend
      TRUE ~ NA_real_),

#check missing values: sum(is.na(ic2024$SL07))  [1] 0

#      adult_perc = case_when(
#        is.na(efage09_7) ~ NA_real_,
#        is.na(efage09_1) ~ NA_real_,
#        TRUE ~ efage09_7 / efage09_1),

#      adult_perc = case_when(
#        is.na(efage09_7) ~ NA_real_,
#        is.na(efage09_1) ~ NA_real_,
#        TRUE ~ efage09_7 / efage09_1),

#      adult = ntile(adult_perc, 4) - 1,                # 0~3: quartiles

#check missing values: sum(is.na(ipeds_learner$EFAGE09_1))  [1] 0
#check missing values: sum(is.na(ipeds_learner$EFAGE09_7))  [1] 9

#      parttime_perc = case_when(
#        is.na(efage06_1) ~ NA_real_,
#        is.na(efage09_1) ~ NA_real_,
#        TRUE ~ efage06_1 / efage09_1),

#      parttime = case_when(
#        is.na(parttime_perc) ~ NA_real_,
#        parttime_perc == 0 ~ 0,                        # none - 7%
#        parttime_perc < 0.25 ~ 1,                     # more than 0%, less than 25% - 8%
#        parttime_perc < 0.50 ~ 2,                     # 25% to less than 50% - 34%
#        parttime_perc <= 1 ~ 3,                       # 50% to 100% - 70%
#        TRUE ~ NA_real_),                             # NA - 14%

#      parttime = ntile(parttime_perc, 4) - 1,          # 0~3: quartiles
#check missing values: sum(is.na(ipeds_learner$EFAGE06_1))  [1] 0

```

```

sports = case_when(
  athassoc == 2 ~ 0,          # 0: not a member
  athassoc == 1 ~ 1,          # 1: member of National Athletic Assn
  TRUE ~ NA_real_),

loan = case_when(
  is.na(loan_a) ~ NA_real_,
  loan_a <= quantile(loan_a, 0.25, na.rm = TRUE) ~ 0,    # 0: lowest
  loan_a <= quantile(loan_a, 0.50, na.rm = TRUE) ~ 1,    # 1: low
  loan_a <= quantile(loan_a, 0.75, na.rm = TRUE) ~ 2,    # 2: high
  TRUE ~ 3),        # 3: highest

```

# Q. Since MLCA requires variables starting at 0, is it appropriate to divide loan amount into

```

asc = case_when(
  carnegiesaec == 2 ~ 0,      # 0: Higher Access, Lower Earnings
  carnegiesaec == 4 ~ 1,      # 1: Higher Access, Medium Earnings
  carnegiesaec == 6 ~ 2,      # 2: Higher Access, Higher Earnings
  carnegiesaec == 1 ~ 3,      # 3: Lower Access, Lower Earnings
  carnegiesaec == 3 ~ 4,      # 4: Lower Access, Medium Earnings
  carnegiesaec == 5 ~ 5,      # 5: Lower Access, Higher Earnings
  TRUE ~ NA_real_),

awardlev = case_when(
  carnegiealf %in% c(1,2) ~ 0, # 0: associates
  carnegiealf == 3 ~ 1,        # 1: baccalaureate
  carnegiealf == 4 ~ 2,        # 2: undergrad / grad-Master's
  carnegiealf == 5 ~ 3,        # 3: undergrad / grad-Doctorate

  is.na(carnegiealf) & (hloffer == 3 | hloffer == 4) ~ 0,
  carnegiealf == -2 & (hloffer == 3 | hloffer == 4) ~ 0,
  hloffer == 5 & instcat == 2 ~ 1,

  TRUE ~ NA_real_),
#Check: x %>% filter(is.na(awardlev)) %>% View()

focus = case_when(
  carnegieapm == -2 ~ 0, # 0: Not classified
  carnegieapm == 1 ~ 1, # 1: Mixed
  carnegieapm == 2 ~ 2, # 2: Professions-focused
  carnegieapm == 3 ~ 3, # 3: Applied and career studies
  carnegieapm %in% c(4,5) ~ 4, # 4: Arts and sciences

```

```

    carnegieapm == 6 ~ 5, # 5: Business
    carnegieapm %in% c(8,9,10) ~ 6, # 6: Medical sciences/health
    carnegieapm == 11 ~ 7, # 7: STEM
    carnegieapm == 12 ~ 8, # 8: Theology
    TRUE ~ NA_real_),

research = case_when(
  carnegiersch == -2 ~ 0, #0: No research designation
  carnegiersch == 3 ~ 1, #1: Research Colleges and Universities
  carnegiersch == 2 ~ 2, #2: High Spending
  carnegiersch == 1 ~ 3, #3: Very High Spending
  TRUE ~ NA_real_),

cbsa = as.character(cbsa),
cbsatype = as.character(cbsatype),

pell = case_when(
  upgrntp < 10 ~ 0,
  upgrntp >= 10 & upgrntp < 30 ~ 1,
  #upgrntp >= 20 & upgrntp < 30 ~ 2,
  upgrntp >= 30 & upgrntp < 50 ~ 2,
  # upgrntp >= 40 & upgrntp < 50 ~ 4,
  upgrntp >= 50 ~ 3,
  TRUE ~ NA_real_)

)

data <- x |>
  filter(distance == 0) %>% # remove observations with only distance e
  group_by(cbsa) %>%
  mutate(inst_num = n_distinct(unitid),
         inst_num = as.numeric(inst_num)) %>%
  ungroup() %>%
  mutate(admrate = admssn/applcn) %>%
  # mutate(admrate = if_else(openadmp == 1, 1, admrate)) |>
  mutate(selectivity = case_when(
    openadmp == 1 ~ 0,
    admrate == 1 ~ 0, #0: open
    admrate >= 0.75 & admrate < 1 ~ 1, #1: less selective
    admrate >= 0.40 & admrate < 0.75 ~ 2, #2: selective
    admrate < 0.40 ~ 3, #3: more selective
  ))

```

```

    TRUE ~ NA_real_),

    i_perc_black = efybkaat/efytotlt,           #Percent of degree-seeking undergrad
    i_perc_white = efywhitt/efytotlt,
    i_perc_hisp = efyhispt/efytotlt,
    i_perc_nhpi = (efyaiant + efynhpit)/efytotlt,
    i_perc_asian = efyasiat/efytotlt,
    i_perc_nonwhite = 1 - i_perc_white,
    nonwhite = ntile(i_perc_nonwhite, 4),
    nonwhite = nonwhite - 1
  ) %>%
rename(region = obereg) %>%
mutate(cbsa = as.character(cbsa))

```

```

z <- read_csv("../data/sources/effy2024.csv") %>%
  filter(EFFYLEV == 2 | EFFYLEV == 4) %>%
  select(UNITID, EFFYALEV, EFYTOTLT) %>%
  mutate(EFFYALEV = if_else(EFFYALEV == 2, "undergrad", "grad")) %>%
  pivot_wider(id_cols = "UNITID",
              names_from = EFFYALEV,
              values_from = EFYTOTLT) %>%
  mutate(total_pop = undergrad + grad,
         per_undergrad = undergrad/total_pop,
         per_grad = grad/total_pop)

```

Rows: 115026 Columns: 72

-- Column specification -----

Delimiter: ","

chr (34): XEYTOTLT, XEYTOTLM, XEYTOTLW, XEFYAIAT, XEFYAIAM, XEFYAIAM, XEFYAS...

dbl (38): UNITID, EFFYALEV, EFFYLEV, LSTUDY, EFYTOTLT, EFYTOTLM, EFYTOTLW, E...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

## -----
## [Census/ACS Data]
## -----

land_area <- core_based_statistical_areas(
  cb = TRUE,
  year = 2024) %>%
  select(GEOID, ALAND)

```

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```
census <- get_acs(geography = "cbsa",
  year = 2024,
  variables = c("B01003_001E", # Total population
    "B19013_001E", # Median Household Income in the Past 12 Months)
```

```

        "DP03_0119PE", # % families below poverty line
        "DP03_0005PE", # % unemployed over 16 years old
        "DP02_0068PE", # % bach or higher
        "B03002_003E", # total white (alone, not hispanic)
        "B03002_004E", # total black (alone, not hispanic)
        "B03002_012E", # total hispanic
        "B03002_006E", # total asian (alone, not hispanic)
        "B03002_001E" # total by ethnicity
    ),
    output = "wide") %>%
left_join(land_area)

```

Getting data from the 2020-2024 5-year ACS

Fetching data by table type ("B/C", "S", "DP") and combining the result.

Joining with `by = join\_by(GEOID)`

```

census <- census |>
  mutate(perc_white = B03002_003E/B03002_001E*100,
         perc_black = B03002_004E/B03002_001E*100,
         perc_hisp = B03002_012E/B03002_001E*100,
         perc_asian = B03002_006E/B03002_001E*100,
         tot_pop = B01003_001E/100000,
         pop_density = B01003_001E/(ALAND/2589988.10)) |> #population per sq mile
  rename(perc_ba = DP02_0068PE,
         med_income = B19013_001E,
         perc_pov = DP03_0119PE,
         perc_unemployed = DP03_0005PE) |>
  select(tot_pop, pop_density, perc_ba,
         med_income, perc_pov, perc_unemployed,
         perc_white, perc_black, perc_hisp, perc_asian,
         GEOID) |>
  mutate(GEOID = as.character(GEOID))

data <- data |>
  left_join(census, by = c("cbsa" = "GEOID"))

```

```

data_mlca <- data %>%
  select(unitid, instnm, stabbr, cbsa, cbsatype, region,
         public, twoyear, size, rurality,
         awardlev,
         religious, focus, housing, research, msi,
         inst_num, selectivity, pell, online, evening, #adult, tuitionnnn, parttime,
         tot_pop, pop_density, perc_ba, med_income, perc_pov, perc_unemployed,
         perc_white, perc_black, perc_hisp, perc_asian,
         i_perc_white, i_perc_black, i_perc_hisp, i_perc_asian, nonwhite) %>%
  as.data.frame()

#metropolitan, tuition,msi, distance,sports,research,

describe(data_mlca)

```

|             | vars | n    | mean      | sd        | median    | trimmed   | mad      |
|-------------|------|------|-----------|-----------|-----------|-----------|----------|
| unitid      | 1    | 3254 | 238590.85 | 120539.08 | 198911.00 | 223531.74 | 59118.67 |
| instnm*     | 2    | 3254 | 1610.98   | 932.84    | 1608.50   | 1610.43   | 1200.16  |
| stabbr*     | 3    | 3254 | 25.69     | 14.59     | 26.00     | 25.72     | 19.27    |
| cbsa*       | 4    | 3254 | 374.03    | 200.50    | 400.00    | 377.35    | 246.11   |
| cbsatype*   | 5    | 3254 | 1.14      | 0.35      | 1.00      | 1.06      | 0.00     |
| region      | 6    | 3254 | 4.49      | 2.12      | 5.00      | 4.44      | 2.97     |
| public      | 7    | 3254 | 0.80      | 0.72      | 1.00      | 0.74      | 1.48     |
| twoyear     | 8    | 3254 | 0.33      | 0.47      | 0.00      | 0.29      | 0.00     |
| size        | 9    | 3254 | 1.20      | 1.18      | 1.00      | 1.05      | 1.48     |
| rurality    | 10   | 3254 | 1.27      | 0.82      | 2.00      | 1.33      | 0.00     |
| awardlev    | 11   | 3232 | 0.99      | 1.12      | 1.00      | 0.87      | 1.48     |
| religious   | 12   | 3254 | 0.21      | 0.41      | 0.00      | 0.14      | 0.00     |
| focus       | 13   | 3254 | 2.74      | 2.04      | 2.00      | 2.46      | 1.48     |
| housing     | 14   | 3251 | 0.55      | 0.50      | 1.00      | 0.56      | 0.00     |
| research    | 15   | 3254 | 0.30      | 0.79      | 0.00      | 0.06      | 0.00     |
| msi         | 16   | 3254 | 0.18      | 0.38      | 0.00      | 0.10      | 0.00     |
| inst_num    | 17   | 3254 | 35.52     | 56.68     | 13.00     | 20.58     | 16.31    |
| selectivity | 18   | 3165 | 0.78      | 0.92      | 0.00      | 0.66      | 0.00     |
| pell        | 19   | 3240 | 1.87      | 0.81      | 2.00      | 1.86      | 1.48     |
| online      | 20   | 3244 | 1.48      | 1.11      | 1.00      | 1.47      | 1.48     |
| evening     | 21   | 3254 | 0.57      | 0.50      | 1.00      | 0.59      | 0.00     |
| tot_pop     | 22   | 3254 | 36.27     | 53.61     | 10.38     | 23.39     | 14.56    |
| pop_density | 23   | 3254 | 743.48    | 897.54    | 359.84    | 544.81    | 390.87   |
| perc_ba     | 24   | 3254 | 35.39     | 9.19      | 36.50     | 35.42     | 8.60     |
| med_income  | 25   | 3254 | 81837.51  | 18132.35  | 79878.50  | 80879.19  | 17562.14 |
| perc_pov    | 26   | 3254 | 8.87      | 2.87      | 8.60      | 8.56      | 2.22     |

|                 |           |           |           |       |       |          |       |
|-----------------|-----------|-----------|-----------|-------|-------|----------|-------|
| perc_unemployed | 27        | 3254      | 3.27      | 0.77  | 3.20  | 3.27     | 0.74  |
| perc_white      | 28        | 3254      | 59.50     | 19.37 | 60.78 | 59.99    | 24.36 |
| perc_black      | 29        | 3254      | 11.87     | 10.32 | 8.29  | 10.29    | 9.33  |
| perc_hisp       | 30        | 3254      | 17.62     | 15.41 | 12.22 | 15.21    | 11.59 |
| perc_asian      | 31        | 3254      | 5.51      | 5.95  | 3.21  | 4.46     | 3.13  |
| i_perc_white    | 32        | 3248      | 0.47      | 0.25  | 0.50  | 0.48     | 0.30  |
| i_perc_black    | 33        | 3248      | 0.17      | 0.20  | 0.10  | 0.13     | 0.10  |
| i_perc_hisp     | 34        | 3248      | 0.18      | 0.18  | 0.12  | 0.15     | 0.11  |
| i_perc_asian    | 35        | 3248      | 0.05      | 0.07  | 0.02  | 0.03     | 0.02  |
| nonwhite        | 36        | 3248      | 1.50      | 1.12  | 1.50  | 1.50     | 1.48  |
|                 |           | min       | max       | range | skew  | kurtosis | se    |
| unitid          | 100654.00 | 500333.00 | 399679.00 | 1.13  | -0.18 | 2113.10  |       |
| instnm*         | 1.00      | 3226.00   | 3225.00   | 0.00  | -1.20 | 16.35    |       |
| stabbr*         | 1.00      | 51.00     | 50.00     | -0.09 | -1.34 | 0.26     |       |
| cbsa*           | 1.00      | 725.00    | 724.00    | -0.20 | -1.15 | 3.51     |       |
| cbsatype*       | 1.00      | 2.00      | 1.00      | 2.02  | 2.09  | 0.01     |       |
| region          | 0.00      | 8.00      | 8.00      | 0.18  | -0.98 | 0.04     |       |
| public          | 0.00      | 2.00      | 2.00      | 0.33  | -1.05 | 0.01     |       |
| twoyear         | 0.00      | 1.00      | 1.00      | 0.70  | -1.50 | 0.01     |       |
| size            | 0.00      | 4.00      | 4.00      | 0.89  | -0.07 | 0.02     |       |
| rurality        | 0.00      | 2.00      | 2.00      | -0.53 | -1.31 | 0.01     |       |
| awardlev        | 0.00      | 3.00      | 3.00      | 0.72  | -0.91 | 0.02     |       |
| religious       | 0.00      | 1.00      | 1.00      | 1.43  | 0.03  | 0.01     |       |
| focus           | 0.00      | 8.00      | 8.00      | 1.13  | 0.30  | 0.04     |       |
| housing         | 0.00      | 1.00      | 1.00      | -0.18 | -1.97 | 0.01     |       |
| research        | 0.00      | 3.00      | 3.00      | 2.61  | 5.50  | 0.01     |       |
| msi             | 0.00      | 1.00      | 1.00      | 1.68  | 0.81  | 0.01     |       |
| inst_num        | 1.00      | 217.00    | 216.00    | 2.33  | 4.46  | 0.99     |       |
| selectivity     | 0.00      | 3.00      | 3.00      | 0.86  | -0.36 | 0.02     |       |
| pell            | 0.00      | 3.00      | 3.00      | 0.05  | -1.08 | 0.01     |       |
| online          | 0.00      | 3.00      | 3.00      | 0.02  | -1.34 | 0.02     |       |
| evening         | 0.00      | 1.00      | 1.00      | -0.28 | -1.92 | 0.01     |       |
| tot_pop         | 0.13      | 197.99    | 197.86    | 1.97  | 3.02  | 0.94     |       |
| pop_density     | 2.61      | 3224.72   | 3222.11   | 1.73  | 1.94  | 15.73    |       |
| perc_ba         | 11.40     | 69.70     | 58.30     | -0.07 | -0.37 | 0.16     |       |
| med_income      | 31354.00  | 162111.00 | 130757.00 | 0.75  | 1.42  | 317.87   |       |
| perc_pov        | 2.80      | 30.50     | 27.70     | 1.76  | 6.42  | 0.05     |       |
| perc_unemployed | 0.40      | 6.60      | 6.20      | 0.13  | -0.01 | 0.01     |       |
| perc_white      | 3.58      | 94.47     | 90.89     | -0.20 | -0.86 | 0.34     |       |
| perc_black      | 0.05      | 74.45     | 74.40     | 1.53  | 3.02  | 0.18     |       |
| perc_hisp       | 0.48      | 95.00     | 94.52     | 1.46  | 2.24  | 0.27     |       |
| perc_asian      | 0.01      | 41.61     | 41.60     | 2.63  | 10.01 | 0.10     |       |
| i_perc_white    | 0.00      | 1.00      | 1.00      | -0.09 | -0.93 | 0.00     |       |

|              |      |      |      |      |       |      |
|--------------|------|------|------|------|-------|------|
| i_perc_black | 0.00 | 1.00 | 1.00 | 2.01 | 3.80  | 0.00 |
| i_perc_hisp  | 0.00 | 1.00 | 1.00 | 1.74 | 2.99  | 0.00 |
| i_perc_asian | 0.00 | 0.95 | 0.95 | 3.64 | 22.04 | 0.00 |
| nonwhite     | 0.00 | 3.00 | 3.00 | 0.00 | -1.36 | 0.02 |

```
x <- data_mlca %>%
  drop_na() %>%
  group_by(cbsa) %>%
  mutate(
    cbsa_id = cur_group_id()
  ) %>%
  as.data.frame()

# data_mlca %>% anti_join(x, by = "unitid") %>% View()

data_mlca <- x

write_rds(data_mlca, "../data/analytic_v2")
```

```
# Variables:
#   Y-      Columns with indicator variables for level 1 class membership (predictors of i
#           # public, two_year, size, msi, award level, research, selectivity

#   iT-     Number of lower-level latent classes

#   id_high- Name of column with higher-level id
#           # CBSA

#   iM-     Number of higher-level latent classes

#   Z-      Columns with lower-level covariates (predictors of level 1 class membership)
#           # rurality, housing, religious, sports

#   Zh-     Columns with higher-level covariates (predictors of level 2 class membership)
#           # Variables- region, number of IHEs, total population, percent bachelors degre

# Level-1 indicators: Observed institution characteristics used to define the latent institution
# Level-1 covariates: Institution-level predictors that explain class membership
# Level-2 covariates: CBSA-level regional variables that explain differences in institution c
```

```

# Lower-level indicators based on framework for insitutional diversity in:
# Harris, M.S. An Empirical Typology of the Institutional Diversity of U.S. Colleges and Uni

#indicators
Y <- c(
  #systemic- structural and governance characteristics
  "public", "size", #"twoyear",

  #programmatic- degree level and academic disciplinary offerings
  "awardlev", "focus", "research",

  #procedural- how programs are offered and implemented (Harris included % of students enrol
    #-the institution / management? operation?
  "online", #ADD VARIABLE FOR THE PERCENTAGE OF STUDENTS WHO ARE EXCLUSIVELY ENROLLED IN ON
    #categorized share of undergraduates enrolled exclusively in distance education
  "evening",
#  "tuitionnn",
#(1)including tuition as quartiles: iT= 8 & iM= 2          (#obs = 1779)
#(2)including tuition & parttime quartiles: iT= 8 & iM= 2    (#obs = 1779)

  #constituentia- "student demographics, background, preparation, and abilities" (Harris, 20
  "selectivity", "pell", "msi", "nonwhite" #, "adult", "parttime"
) #level-1 indicators

Z <- c("i_perc_white", "i_perc_black", "i_perc_hisp", "i_perc_asian", "housing", "rurality",

Zh <- c("cbsatype", "inst_num", "tot_pop", "pop_density", "perc_ba", "med_income", "perc_pov

## -----
## Descriptive Statistics
## -----

# -----
# Level 1: Institution-level variables
# -----

# Categorical Level-1 variables
l1_cat <- c("public", "size", "awardlev", "focus", "research", "online",
  "evening", "selectivity", "pell", "msi", "nonwhite",
  "housing", "rurality", "religious")

```

```
# Continuous Level-1 variables
l1_cont <- c("i_perc_white", "i_perc_black", "i_perc_hisp", "i_perc_asian")

# Frequency and percentage for categorical variables
l1_categorical_desc <- data_mlca |>
  select(all_of(l1_cat)) |>
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Category") |>
  count(Variable, Category, name = "N") |>
  group_by(Variable) |>
  mutate(Percent = round(100 * N / sum(N), 2)) |>
  ungroup()

l1_categorical_desc
```

```
# A tibble: 52 x 4
  Variable Category      N Percent
  <chr>         <dbl> <int>   <dbl>
1 awardlev         0  1471   47.0
2 awardlev         1   759   24.2
3 awardlev         2   392   12.5
4 awardlev         3   511   16.3
5 evening          0  1335   42.6
6 evening          1  1798   57.4
7 focus            0    59    1.88
8 focus            1   933   29.8
9 focus            2  1037   33.1
10 focus           3   291    9.29
# i 42 more rows
```

```
# Descriptive statistics for continuous variables
l1_continuous_desc <- data_mlca |>
  summarise(across(all_of(l1_cont),
                    list(N = ~sum(!is.na(.)),
                         Mean = ~mean(., na.rm = TRUE),
                         SD = ~sd(., na.rm = TRUE),
                         Min = ~min(., na.rm = TRUE),
                         Max = ~max(., na.rm = TRUE)))) |>
  pivot_longer(everything(),
```

```

      names_to = c("Variable", ".value"),
      names_pattern = "(.+)_ (N|Mean|SD|Min|Max)$" ) |>
mutate(across(c(Mean, SD, Min, Max), ~round(., 3)))

l1_continuous_desc

```

```

# A tibble: 4 x 6
  Variable      N Mean  SD   Min  Max
  <chr>      <int> <dbl> <dbl> <dbl> <dbl>
1 i_perc_white 3133 0.475 0.253    0    1
2 i_perc_black 3133 0.173 0.2     0    1
3 i_perc_hisp  3133 0.181 0.181    0    1
4 i_perc_asian 3133 0.045 0.064    0 0.623

```

```

# -----
# Level 2: CBSA-level variables
# -----

# Keep one observation per CBSA
cbsa_data <- data_mlca |>
  distinct(cbsa_id, .keep_all = TRUE)

# Categorical Level-2 variable
l2_categorical_desc <- cbsa_data |>
  count(cbsatype, name = "N") |>
  mutate(Percent = round(100 * N / sum(N), 2))

l2_categorical_desc

```

```

  cbsatype    N Percent
1         1 375   51.72
2         2 350   48.28

```

```

# Continuous Level-2 variables
l2_cont <- c("inst_num", "tot_pop", "pop_density", "perc_ba", "med_income", "perc_pov",
            "perc_unemployed", "perc_white", "perc_black", "perc_hisp", "perc_asian")

l2_continuous_desc <- cbsa_data |>
  summarise(across(
    all_of(l2_cont),

```

```

list(N = ~sum(!is.na(.)),
     Mean = ~mean(., na.rm = TRUE),
     SD = ~sd(., na.rm = TRUE),
     Min = ~min(., na.rm = TRUE),
     Max = ~max(., na.rm = TRUE))) |>
pivot_longer(everything(),
              names_to = c("Variable", ".value"),
              names_pattern = "(.+)_ (N|Mean|SD|Min|Max)$") |>
mutate(across(c(Mean, SD, Min, Max), ~round(., 2)))

l2_continuous_desc

```

# A tibble: 11 x 6

|    | Variable<br><chr> | N<br><int> | Mean<br><dbl> | SD<br><dbl> | Min<br><dbl> | Max<br><dbl> |
|----|-------------------|------------|---------------|-------------|--------------|--------------|
| 1  | inst_num          | 725        | 4.49          | 11.8        | 1            | 217          |
| 2  | tot_pop           | 725        | 4.24          | 12.6        | 0.13         | 198.         |
| 3  | pop_density       | 725        | 191.          | 280.        | 2.61         | 3225.        |
| 4  | perc_ba           | 725        | 28.6          | 9.29        | 11.4         | 69.7         |
| 5  | med_income        | 725        | 69572.        | 14640.      | 31354        | 162111       |
| 6  | perc_pov          | 725        | 9.82          | 3.81        | 2.8          | 30.5         |
| 7  | perc_unemployed   | 725        | 2.92          | 0.84        | 0.4          | 6.6          |
| 8  | perc_white        | 725        | 70.0          | 18.6        | 3.58         | 94.5         |
| 9  | perc_black        | 725        | 8.96          | 11.8        | 0.05         | 74.4         |
| 10 | perc_hisp         | 725        | 13.0          | 15.1        | 0.48         | 95           |
| 11 | perc_asian        | 725        | 2.29          | 3.53        | 0.01         | 41.6         |

```

# -----
# Sample sizes
# -----

cat("Number of institutions:", nrow(data_mlca), "\n")

```

Number of institutions: 3133

```

cat("Number of CBSAs:", nrow(cbsa_data), "\n")

```

Number of CBSAs: 725

```
# -----
# Save descriptive statistics
# -----

sample_sizes <- tibble(
  Level = c("Institutions", "CBSAs"),
  N = c(nrow(data_mlca), nrow(cbsa_data)))

write_xlsx(
  list("L1 categorical" = l1_categorical_desc,
       "L1 continuous" = l1_continuous_desc,
       "L2 categorical" = l2_categorical_desc,
       "L2 continuous" = l2_continuous_desc,
       "Sample sizes" = sample_sizes),
  "../data/MLCA_descriptive_statistics.xlsx")
```

```
# Level-1 indicators (Y)
## cf. these are not predictors, so technically we don't need to check for multicollinearity
## cf. these are categorical variables, so VIF may not be appropriate
y_data <- data_mlca |> select(all_of(Y))

set.seed(1234)
y_data$dummy_y <- rnorm(nrow(y_data))

vif_y_model <- lm(dummy_y ~ ., data = y_data)
vif(vif_y_model)
```

| public   | size        | awardlev | focus    | research | online   |
|----------|-------------|----------|----------|----------|----------|
| 1.760092 | 2.488298    | 2.215250 | 1.349799 | 2.272521 | 1.397997 |
| evening  | selectivity | pell     | msi      | nonwhite |          |
| 1.132332 | 1.959964    | 1.438918 | 1.491228 | 1.616924 |          |

```
# Level-1 covariates (Z)
z_data <- data_mlca |>
  select(all_of(Z))

set.seed(1234)
z_data$dummy_y <- rnorm(nrow(z_data))

vif_z_model <- lm(dummy_y ~ ., data = z_data)
vif(vif_z_model)
```

|              |              |             |              |          |          |
|--------------|--------------|-------------|--------------|----------|----------|
| i_perc_white | i_perc_black | i_perc_hisp | i_perc_asian | housing  | rurality |
| 5.605996     | 4.024443     | 3.505072    | 1.664504     | 1.347178 | 1.113060 |
| religious    |              |             |              |          |          |
| 1.213664     |              |             |              |          |          |

```
# Level-2 covariates (Zh)
zh_data <- data_mlca |>
  select(all_of(Zh)) |>
  mutate(cbsatype = as.factor(cbsatype)) # switch nominal variable to factor

zh_data$dummy_y <- rnorm(nrow(zh_data))

vif_zh_model <- lm(dummy_y ~ ., data = zh_data)
vif(vif_zh_model)
```

|            |            |                 |             |            |
|------------|------------|-----------------|-------------|------------|
| cbsatype   | inst_num   | tot_pop         | pop_density | perc_ba    |
| 1.489369   | 55.185840  | 51.956170       | 14.507700   | 3.756909   |
| med_income | perc_pov   | perc_unemployed | perc_white  | perc_black |
| 7.506924   | 4.332139   | 2.383692        | 31.438482   | 9.009204   |
| perc_hisp  | perc_asian |                 |             |            |
| 19.233788  | 8.332143   |                 |             |            |

```
set.seed(1234)

model_selection <- multiLCA(
  data = data_mlca,
  Y = Y,
  iT = 6:15,
  Z = Z,
  iM = 2:8,
  id_high = c("cbsa_id"),
  Zh = Zh,
  incomplete = FALSE,
  extout = TRUE,
  verbose = TRUE,
  sequential = TRUE,
  reord = FALSE
)
```

Fitting single-level measurement model (step 1)...

Fitting single-level measurement model (step 1)...

Fitting single-level measurement model (step 1)...

Fitting single-level measurement model (step 1)...

Fitting single-level measurement model (step 1)...

Fitting single-level measurement model (step 1)...

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Fitting single-level measurement model (step 1)...

Fitting multilevel measurement model (step 2)...

Fitting multilevel measurement model (step 2)...

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Fitting multilevel measurement model (step 3)...

Fitting multilevel measurement model (step 3)...

Fitting multilevel measurement model (step 3)...

Fitting multilevel measurement model (step 3)...

Fitting multilevel measurement model (step 3)...

\$step1

|       | BIClow   | BIChigh  | AIC      | ICL_BIClow | ICL_BIChigh |
|-------|----------|----------|----------|------------|-------------|
| iT=6  | 62283.9  | 61978.02 | 61019.51 | -          | -           |
| iT=7  | 62168.61 | 61811.5  | 60692.47 | -          | -           |
| iT=8  | 61791.57 | 61383.23 | 60103.69 | -          | -           |
| iT=9  | 61450.78 | 60991.22 | 59551.16 | -          | -           |
| iT=10 | 61378.01 | 60867.22 | 59266.65 | -          | -           |
| iT=11 | 61477.86 | 60915.85 | 59154.76 | -          | -           |
| iT=12 | 61356.19 | 60742.95 | 58821.34 | -          | -           |
| iT=13 | 61244.3  | 60579.83 | 58497.71 | -          | -           |
| iT=14 | 61325.93 | 60610.24 | 58367.6  | -          | -           |
| iT=15 | 61436.13 | 60669.22 | 58266.07 | -          | -           |

\$step2

|          | BIClow   | BIChigh  | AIC      | ICL_BIClow | ICL_BIChigh |
|----------|----------|----------|----------|------------|-------------|
| iT*,iM=2 | 60867.33 | 60183.84 | 58042.1  | 61910.42   | 60417.86    |
| iT*,iM=3 | 60979.2  | 60276.68 | 58075.32 | 62316.66   | 60970.78    |
| iT*,iM=4 | 60751.79 | 60030.25 | 57769.27 | 62135.21   | 61024.22    |

|          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| iT*,iM=5 | 60803.87 | 60063.3  | 57742.7  | 62169.53 | 61249.8  |
| iT*,iM=6 | 60713.77 | 59954.18 | 57573.95 | 62056.67 | 61233.75 |
| iT*,iM=7 | 61027.61 | 60248.98 | 57809.14 | 62274.47 | 61597.22 |
| iT*,iM=8 | 60930.15 | 60132.5  | 57633.04 | 62272.32 | 61274.15 |

\$step3

|           | BIClow   | BIChigh  | AIC      | ICL_BIClow | ICL_BIChigh |
|-----------|----------|----------|----------|------------|-------------|
| iT=6,iM*  | 62444.77 | 62094.97 | 60998.88 | 63138.45   | 63550.71    |
| iT=7,iM*  | 61352.09 | 60943.76 | 59664.22 | 62103.78   | 61952.13    |
| iT=8,iM*  | 60958.47 | 60491.59 | 59028.6  | 61770.99   | 61548.56    |
| iT=9,iM*  | 60836.74 | 60311.32 | 58664.88 | 61645.79   | 61575.61    |
| iT=10,iM* | 60761.1  | 60177.14 | 58347.25 | 61716.34   | 61314.49    |
| iT=11,iM* | 60698.88 | 60056.37 | 58043.04 | 61721.76   | 61138.39    |
| iT=12,iM* | 60682.6  | 59981.54 | 57784.77 | 61820.49   | 60968.09    |
| iT=13,iM* | 60920.58 | 60160.99 | 57780.77 | 62278.63   | 61205.84    |
| iT=14,iM* | 61034.27 | 60216.13 | 57652.46 | 62335.95   | 61333.29    |
| iT=15,iM* | 61041.56 | 60164.88 | 57417.76 | 62569.08   | 61132.8     |

\$optimal

iT= 12  
iM= 6

```
analytic_model <- model_selection
```

```
analytic_model
```

CALL:

```
multiLCA(data = data_mlca, Y = Y, iT = 6:15, id_high = c("cbsa_id"),
  iM = 2:8, Z = Z, Zh = Zh, incomplete = FALSE, extout = TRUE,
  sequential = TRUE, reord = FALSE, verbose = TRUE)
```

SPECIFICATION:

Multilevel LC model

MISSING VALUES ON THE INDICATORS:

None

FINAL SAMPLE SIZE:

3133

ESTIMATION DETAILS:

EMiter    LLfirst    LLlast  
838 -28932.99 -28413.38

-----  
GROUP PROPORTIONS:

P(G1) 0.0032  
P(G2) 0.4463  
P(G3) 0.0388  
P(G4) 0.3274  
P(G5) 0.0872  
P(G6) 0.0971

CLASS PROPORTIONS:

|          | G1     | G2     | G3     | G4     | G5     | G6     |
|----------|--------|--------|--------|--------|--------|--------|
| P(C1 G)  | 0.0845 | 0.0050 | 0.1022 | 0.0959 | 0.1506 | 0.0000 |
| P(C2 G)  | 0.3799 | 0.0520 | 0.0000 | 0.0275 | 0.0146 | 0.0203 |
| P(C3 G)  | 0.1558 | 0.0000 | 0.0890 | 0.0339 | 0.1179 | 0.2248 |
| P(C4 G)  | 0.0522 | 0.0558 | 0.0636 | 0.0688 | 0.0338 | 0.0045 |
| P(C5 G)  | 0.0416 | 0.1104 | 0.0000 | 0.1664 | 0.0245 | 0.0000 |
| P(C6 G)  | 0.0924 | 0.0083 | 0.0000 | 0.0000 | 0.2661 | 0.0647 |
| P(C7 G)  | 0.0122 | 0.2046 | 0.0455 | 0.1017 | 0.0073 | 0.0000 |
| P(C8 G)  | 0.0077 | 0.0180 | 0.2511 | 0.2038 | 0.0000 | 0.5303 |
| P(C9 G)  | 0.0718 | 0.0248 | 0.0812 | 0.0784 | 0.0793 | 0.0000 |
| P(C10 G) | 0.0054 | 0.1187 | 0.0857 | 0.0670 | 0.0000 | 0.1553 |
| P(C11 G) | 0.0887 | 0.0806 | 0.2818 | 0.1156 | 0.3061 | 0.0000 |
| P(C12 G) | 0.0079 | 0.3218 | 0.0000 | 0.0411 | 0.0000 | 0.0000 |

RESPONSE PROBABILITIES:

|               | C1     | C2     | C3     | C4     | C5     | C6     | C7     | C8     |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|
| P(public.0 C) | 0.5877 | 0.9848 | 0.4106 | 0.9438 | 1.0000 | 0.0045 | 0.6991 | 0.0000 |
| P(public.1 C) | 0.0059 | 0.0152 | 0.5894 | 0.0284 | 0.0000 | 0.9955 | 0.3009 | 1.0000 |
| P(public.2 C) | 0.4065 | 0.0000 | 0.0000 | 0.0278 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| P(size.0 C)   | 0.6788 | 0.9804 | 0.0511 | 0.2326 | 0.0741 | 0.0403 | 0.3685 | 0.0292 |
| P(size.1 C)   | 0.2248 | 0.0196 | 0.4389 | 0.7565 | 0.7674 | 0.1334 | 0.6148 | 0.4450 |
| P(size.2 C)   | 0.0468 | 0.0000 | 0.2787 | 0.0110 | 0.1300 | 0.3281 | 0.0134 | 0.3576 |

|                    |        |        |        |        |        |        |        |        |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| P(size.3 C)        | 0.0171 | 0.0000 | 0.1432 | 0.0000 | 0.0203 | 0.3385 | 0.0000 | 0.1399 |
| P(size.4 C)        | 0.0326 | 0.0000 | 0.0880 | 0.0000 | 0.0082 | 0.1597 | 0.0033 | 0.0284 |
| P(awardlev.0 C)    | 0.3006 | 0.3452 | 0.0151 | 0.0326 | 0.0000 | 0.9909 | 0.0603 | 0.9697 |
| P(awardlev.1 C)    | 0.4775 | 0.6519 | 0.2855 | 0.8668 | 0.0093 | 0.0000 | 0.9397 | 0.0124 |
| P(awardlev.2 C)    | 0.1387 | 0.0029 | 0.3865 | 0.1005 | 0.4681 | 0.0000 | 0.0000 | 0.0125 |
| P(awardlev.3 C)    | 0.0832 | 0.0000 | 0.3129 | 0.0000 | 0.5226 | 0.0091 | 0.0000 | 0.0054 |
| P(focus.0 C)       | 0.0000 | 0.0398 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0175 | 0.0000 |
| P(focus.1 C)       | 0.0538 | 0.0180 | 0.4211 | 0.1237 | 0.1468 | 0.8614 | 0.1570 | 0.5173 |
| P(focus.2 C)       | 0.3030 | 0.0416 | 0.4702 | 0.0000 | 0.7072 | 0.1342 | 0.5475 | 0.4090 |
| P(focus.3 C)       | 0.0355 | 0.0573 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0098 | 0.0674 |
| P(focus.4 C)       | 0.1249 | 0.0092 | 0.0950 | 0.8381 | 0.0000 | 0.0000 | 0.1926 | 0.0000 |
| P(focus.5 C)       | 0.1422 | 0.0069 | 0.0050 | 0.0051 | 0.0334 | 0.0000 | 0.0060 | 0.0000 |
| P(focus.6 C)       | 0.2533 | 0.0673 | 0.0087 | 0.0000 | 0.0822 | 0.0043 | 0.0578 | 0.0063 |
| P(focus.7 C)       | 0.0076 | 0.0103 | 0.0000 | 0.0331 | 0.0096 | 0.0000 | 0.0118 | 0.0000 |
| P(focus.8 C)       | 0.0797 | 0.7496 | 0.0000 | 0.0000 | 0.0208 | 0.0000 | 0.0000 | 0.0000 |
| P(research.0 C)    | 1.0000 | 1.0000 | 0.5481 | 0.7979 | 0.9404 | 0.9949 | 0.9662 | 1.0000 |
| P(research.1 C)    | 0.0000 | 0.0000 | 0.2576 | 0.2021 | 0.0490 | 0.0051 | 0.0338 | 0.0000 |
| P(research.2 C)    | 0.0000 | 0.0000 | 0.1686 | 0.0000 | 0.0106 | 0.0000 | 0.0000 | 0.0000 |
| P(research.3 C)    | 0.0000 | 0.0000 | 0.0258 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| P(online.0 C)      | 0.1771 | 0.7750 | 0.0363 | 0.8636 | 0.0891 | 0.0046 | 0.1444 | 0.0078 |
| P(online.1 C)      | 0.2401 | 0.0415 | 0.4944 | 0.1010 | 0.4875 | 0.0577 | 0.5276 | 0.0340 |
| P(online.2 C)      | 0.1193 | 0.0944 | 0.3265 | 0.0354 | 0.3005 | 0.4126 | 0.2056 | 0.3322 |
| P(online.3 C)      | 0.4635 | 0.0891 | 0.1428 | 0.0000 | 0.1228 | 0.5250 | 0.1223 | 0.6260 |
| P(evening C)       | 0.5875 | 0.1263 | 0.6634 | 0.0135 | 0.6295 | 0.7736 | 0.4595 | 0.7381 |
| P(selectivity.0 C) | 0.6424 | 0.4263 | 0.0480 | 0.0000 | 0.0331 | 1.0000 | 0.0478 | 0.9936 |
| P(selectivity.1 C) | 0.1978 | 0.2898 | 0.5986 | 0.2620 | 0.5586 | 0.0000 | 0.5743 | 0.0064 |
| P(selectivity.2 C) | 0.1262 | 0.2121 | 0.3245 | 0.3990 | 0.3882 | 0.0000 | 0.3422 | 0.0000 |
| P(selectivity.3 C) | 0.0335 | 0.0718 | 0.0289 | 0.3390 | 0.0200 | 0.0000 | 0.0356 | 0.0000 |
| P(pell.0 C)        | 0.0366 | 0.0622 | 0.0000 | 0.0410 | 0.0093 | 0.0195 | 0.0001 | 0.0057 |
| P(pell.1 C)        | 0.0743 | 0.1696 | 0.0500 | 0.7639 | 0.4197 | 0.6376 | 0.3301 | 0.3948 |
| P(pell.2 C)        | 0.2976 | 0.3551 | 0.4437 | 0.1478 | 0.5268 | 0.2922 | 0.6172 | 0.4937 |
| P(pell.3 C)        | 0.5915 | 0.4131 | 0.5063 | 0.0473 | 0.0441 | 0.0506 | 0.0526 | 0.1058 |
| P(msi C)           | 0.2734 | 0.0000 | 0.8884 | 0.0406 | 0.0198 | 0.8481 | 0.0293 | 0.2191 |
| P(nonwhite.0 C)    | 0.0000 | 0.8801 | 0.0000 | 0.2864 | 0.3353 | 0.0000 | 0.4323 | 0.0000 |
| P(nonwhite.1 C)    | 0.0925 | 0.0797 | 0.0219 | 0.4119 | 0.4663 | 0.0691 | 0.4088 | 0.3624 |
| P(nonwhite.2 C)    | 0.4132 | 0.0090 | 0.3056 | 0.2277 | 0.1870 | 0.3437 | 0.1533 | 0.5066 |
| P(nonwhite.3 C)    | 0.4943 | 0.0312 | 0.6725 | 0.0740 | 0.0114 | 0.5872 | 0.0056 | 0.1310 |
|                    | C9     | C10    | C11    | C12    |        |        |        |        |
| P(public.0 C)      | 0.4141 | 0.0375 | 0.0918 | 0.0063 |        |        |        |        |
| P(public.1 C)      | 0.5859 | 0.9625 | 0.0037 | 0.9937 |        |        |        |        |
| P(public.2 C)      | 0.0000 | 0.0000 | 0.9046 | 0.0000 |        |        |        |        |
| P(size.0 C)        | 0.0000 | 0.0000 | 0.8542 | 0.0467 |        |        |        |        |
| P(size.1 C)        | 0.0220 | 0.1274 | 0.1397 | 0.7295 |        |        |        |        |

|                    |        |        |        |        |
|--------------------|--------|--------|--------|--------|
| P(size.2 C)        | 0.1639 | 0.3652 | 0.0045 | 0.1809 |
| P(size.3 C)        | 0.2674 | 0.4002 | 0.0017 | 0.0412 |
| P(size.4 C)        | 0.5467 | 0.1071 | 0.0000 | 0.0018 |
| P(awardlev.0 C)    | 0.0000 | 0.0000 | 0.9471 | 0.9703 |
| P(awardlev.1 C)    | 0.0000 | 0.0706 | 0.0529 | 0.0297 |
| P(awardlev.2 C)    | 0.0042 | 0.5812 | 0.0000 | 0.0000 |
| P(awardlev.3 C)    | 0.9958 | 0.3483 | 0.0000 | 0.0000 |
| P(focus.0 C)       | 0.0000 | 0.0000 | 0.0974 | 0.0000 |
| P(focus.1 C)       | 0.7742 | 0.3360 | 0.0049 | 0.4160 |
| P(focus.2 C)       | 0.1479 | 0.6447 | 0.0390 | 0.4802 |
| P(focus.3 C)       | 0.0000 | 0.0000 | 0.4717 | 0.0564 |
| P(focus.4 C)       | 0.0046 | 0.0126 | 0.0161 | 0.0023 |
| P(focus.5 C)       | 0.0000 | 0.0000 | 0.0553 | 0.0032 |
| P(focus.6 C)       | 0.0064 | 0.0000 | 0.3156 | 0.0419 |
| P(focus.7 C)       | 0.0668 | 0.0066 | 0.0000 | 0.0000 |
| P(focus.8 C)       | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| P(research.0 C)    | 0.0000 | 0.4098 | 1.0000 | 1.0000 |
| P(research.1 C)    | 0.0422 | 0.2626 | 0.0000 | 0.0000 |
| P(research.2 C)    | 0.1678 | 0.2448 | 0.0000 | 0.0000 |
| P(research.3 C)    | 0.7900 | 0.0828 | 0.0000 | 0.0000 |
| P(online.0 C)      | 0.1128 | 0.0000 | 0.7306 | 0.0000 |
| P(online.1 C)      | 0.7116 | 0.4111 | 0.1028 | 0.0803 |
| P(online.2 C)      | 0.1594 | 0.4775 | 0.0604 | 0.5888 |
| P(online.3 C)      | 0.0163 | 0.1114 | 0.1062 | 0.3309 |
| P(evening C)       | 0.5658 | 0.5041 | 0.7092 | 0.6027 |
| P(selectivity.0 C) | 0.0000 | 0.0373 | 0.8640 | 0.9914 |
| P(selectivity.1 C) | 0.3200 | 0.7584 | 0.0874 | 0.0000 |
| P(selectivity.2 C) | 0.4032 | 0.1971 | 0.0334 | 0.0086 |
| P(selectivity.3 C) | 0.2768 | 0.0072 | 0.0152 | 0.0000 |
| P(pell.0 C)        | 0.0000 | 0.0000 | 0.0121 | 0.0067 |
| P(pell.1 C)        | 0.7507 | 0.4503 | 0.0539 | 0.5061 |
| P(pell.2 C)        | 0.2417 | 0.5398 | 0.1525 | 0.4814 |
| P(pell.3 C)        | 0.0076 | 0.0099 | 0.7816 | 0.0058 |
| P(msi C)           | 0.1415 | 0.0018 | 0.0072 | 0.0028 |
| P(nonwhite.0 C)    | 0.1618 | 0.5062 | 0.0858 | 0.6592 |
| P(nonwhite.1 C)    | 0.2928 | 0.4026 | 0.0806 | 0.3408 |
| P(nonwhite.2 C)    | 0.4317 | 0.0912 | 0.2499 | 0.0000 |
| P(nonwhite.3 C)    | 0.1138 | 0.0000 | 0.5837 | 0.0000 |

-----

MODEL AND CLASSIFICATION STATISTICS:

```

R2entrlow      0.9254
R2entrhigh     0.4803
BIClow         60682.5972
BIChigh        59981.5449
ICLBIClow      61820.4883
ICLBIChigh     60968.0876
AIC            57784.7687

```

“{r}

```
set.seed(1234)
```

### testing 8 lower-level classes, 2 higher-level clusters

```
test_l8_h2<- multiLCA( data = data_mlca, Y = Y, iT = 8, Z = Z, iM = 2, id_high =
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
= TRUE, reord = FALSE )
```

### testing 9 lower-level classes, 2 higher-level clusters

```
test_l9_h2<- multiLCA( data = data_mlca, Y = Y, iT = 9, Z = Z, iM = 2, id_high =
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
= TRUE, reord = FALSE )
```

### testing 10 lower-level classes, 2 higher-level clusters

```
test_l10_h2<- multiLCA( data = data_mlca, Y = Y, iT = 10, Z = Z, iM = 2, id_high =
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
= TRUE, reord = FALSE )
```

### testing 11 lower-level classes, 2 higher-level clusters

```
test_l11_h2<- multiLCA( data = data_mlca, Y = Y, iT = 11, Z = Z, iM = 2, id_high =
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
= TRUE, reord = FALSE )
```

### **testing 8 lower-level classes, 3 higher-level clusters**

```
test_l8_h3<- multiLCA( data = data_mlca, Y = Y, iT = 8, Z = Z, iM = 3, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

### **testing 9 lower-level classes, 3 higher-level clusters**

```
test_l9_h3<- multiLCA( data = data_mlca, Y = Y, iT = 9, Z = Z, iM = 3, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```



**NO SOLUTION FOUND**

**# testing 10 lower-level classes, 3 higher-level clusters**

```
test_l10_h3<- multiLCA(
```

```
data = data_mlca,
```

```
Y = Y,
```

```
iT = 10,
```

```
Z = Z,
```

```
iM = 3,
```

```
id_high = c("cbsa_id"),
```

```
Zh = Zh,
```

```
incomplete = FALSE,
```

```
extout = TRUE,
```

```
verbose = TRUE,
```

```
sequential = TRUE,
```

```
reord = FALSE
```

```
)
```

**testing 11 lower-level classes, 3 higher-level clusters**

```
test_l11_h3 <- multiLCA( data = data_mlca, Y = Y, iT = 11, Z = Z, iM = 3, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
```

```
= TRUE, reord = FALSE )
```

### **testing 8 lower-level classes, 4 higher-level clusters**

```
test_l8_h4 <- multiLCA( data = data_mlca, Y = Y, iT = 8, Z = Z, iM = 4, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```



**NO SOLUTION FOUND**

**# testing 9 lower-level classes, 4 higher-level clusters**

```
test_l9_h4<- multiLCA(  
  data = data_mlca,  
  Y = Y,  
  iT = 9,  
  Z = Z,  
  iM = 4,  
  id_high = c("cbsa_id"),  
  Zh = Zh,  
  incomplete = FALSE,  
  extout = TRUE,  
  verbose = TRUE,  
  sequential = TRUE,  
  reord = FALSE  
)
```

**testing 10 lower-level classes, 4 higher-level clusters**

```
test_l10_h4<- multiLCA( data = data_mlca, Y = Y, iT = 10, Z = Z, iM = 4, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
```

= TRUE, reord = FALSE )



**NO SOLUTION FOUND**

**# testing 11 lower-level classes, 4 higher-level clusters**

**test\_l11\_h4<- multiLCA(**

**data = data\_mlca,**

**Y = Y,**

**iT = 11,**

**Z = Z,**

**iM = 4,**

**id\_high = c("cbsa\_id"),**

**Zh = Zh,**

**incomplete = FALSE,**

**extout = TRUE,**

**verbose = TRUE,**

**sequential = TRUE,**

**reord = FALSE**

**)**

**NO SOLUTION FOUND**

**# testing 8 lower-level classes, 5 higher-level clusters**

**test\_l8\_h5 <- multiLCA(**

**data = data\_mlca,**

**Y = Y,**

```

= TRUE, reord = FALSE )
test_l8_h2 test_l9_h2 test_l10_h2 test_l11_h2
test_l8_h3 test_l9_h3 #test_l10_h3 test_l11_h3
test_l8_h4 #test_l9_h4 test_l10_h4 #test_l11_h4
#test_l8_h5 #test_l9_h5 #test_l10_h5 test_l11_h5

```

```

# ``{r}

```

```

set.seed(1234)

```

```

#Single-level LCA without covariates

```

```

test_1 <- multiLCA(
  data = data_mlca,
  Y      = Y,
  iT     = 6:12,
  # Z     = Z,
  # iM    = 2:5,
  # id_high = c("cbsa_id"),
  # Zh    = Zh,
  incomplete = FALSE,
  extout     = TRUE,
  verbose    = TRUE,
  sequential = TRUE,
  reord      = FALSE
)

```

```

#Single-level LCA with covariates

```

```

test_1c <- multiLCA(
  data = data_mlca,
  Y      = Y,
  iT     = 6:12,
  Z      = Z,
  # iM    = 2:5,
  # id_high = c("cbsa_id"),
  # Zh    = Zh,
  incomplete = FALSE,
  extout     = TRUE,

```

```

    verbose      = TRUE,
    sequential    = TRUE,
    reord         = FALSE
)

```

```

test_1
test_1c

```

```

summary(test_1)
summary(test_1c)

```

```

#Full multilevel analysis with covariates

```

```

final_model <- multiLCA(
  data = data_mlca,
  Y      = Y,
  iT     = 6:12,
  Z      = Z,
  iM     = 2:5,
  id_high = c("cbsa_id"),
  Zh     = Zh,
  incomplete = FALSE,
  extout  = TRUE,
  verbose = TRUE,
  sequential = TRUE,
  reord   = FALSE
)

```

```

final_model
summary(final_model)
plot(final_model)

```

```

l1_proportions <- as.data.frame(analytic_model$mPi) # Lower-level class proportions given t
l1_proportions

```

|         | G1          | G2          | G3           | G4           | G5           |
|---------|-------------|-------------|--------------|--------------|--------------|
| P(C1 G) | 0.084469305 | 0.005029669 | 1.021915e-01 | 9.593473e-02 | 1.505628e-01 |

|          |             |             |               |               |              |
|----------|-------------|-------------|---------------|---------------|--------------|
| P(C2 G)  | 0.379918307 | 0.052027929 | 0.000000e+00  | 2.753982e-02  | 1.455763e-02 |
| P(C3 G)  | 0.155805080 | 0.000000000 | 8.895156e-02  | 3.385052e-02  | 1.179301e-01 |
| P(C4 G)  | 0.052178660 | 0.055827976 | 6.358374e-02  | 6.881983e-02  | 3.376384e-02 |
| P(C5 G)  | 0.041568758 | 0.110350326 | 0.000000e+00  | 1.663651e-01  | 2.447481e-02 |
| P(C6 G)  | 0.092363237 | 0.008307892 | 0.000000e+00  | 2.716762e-130 | 2.660715e-01 |
| P(C7 G)  | 0.012195497 | 0.204573285 | 4.550189e-02  | 1.016558e-01  | 7.258132e-03 |
| P(C8 G)  | 0.007682066 | 0.017958858 | 2.510565e-01  | 2.038252e-01  | 6.086319e-40 |
| P(C9 G)  | 0.071787219 | 0.024761026 | 8.120281e-02  | 7.835041e-02  | 7.928151e-02 |
| P(C10 G) | 0.005404718 | 0.118707282 | 8.572620e-02  | 6.695188e-02  | 0.000000e+00 |
| P(C11 G) | 0.088718997 | 0.080623389 | 2.817858e-01  | 1.155717e-01  | 3.060996e-01 |
| P(C12 G) | 0.007908156 | 0.321832367 | 9.554381e-237 | 4.113501e-02  | 0.000000e+00 |

G6

|          |               |
|----------|---------------|
| P(C1 G)  | 0.000000e+00  |
| P(C2 G)  | 2.034539e-02  |
| P(C3 G)  | 2.248333e-01  |
| P(C4 G)  | 4.506883e-03  |
| P(C5 G)  | 5.077412e-252 |
| P(C6 G)  | 6.473113e-02  |
| P(C7 G)  | 0.000000e+00  |
| P(C8 G)  | 5.302828e-01  |
| P(C9 G)  | 0.000000e+00  |
| P(C10 G) | 1.553005e-01  |
| P(C11 G) | 1.551609e-15  |
| P(C12 G) | 4.951244e-42  |

```
l1_probabilities <- as.data.frame(analytic_model$mPhi)
```

```
l2_posterior <- as.data.frame(analytic_model$mPW)
```

```
l2_classes <- data.frame(
  group_id = row.names(analytic_model$mPW),
  L2_Predicted_Class = apply(l2_posterior, 1, which.max),
  L2_Prob_Class1 = l2_posterior[, 1],
  L2_Prob_Class2 = l2_posterior[, 2],
  L2_Prob_Class3 = l2_posterior[, 3]
)
```

```
l2_class_data <- data_mlca %>%
  mutate(cbsa_id = as.character(cbsa_id)) %>%
  left_join(l2_classes,
```

```
      by = c("cbasa_id" = "group_id"))

l2_class_summary <- l2_class_data %>%
  select(cbasa, L2_Predicted_Class ) %>%
  distinct()

inst_cbasa_classification <- l2_class_data

write_csv(inst_cbasa_classification, "../data_paper2/inst_cbasa_classification.csv")
```